

Yukti Gupta

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Objective: Seeking a Software Developer position where I can leverage skills in Python, Java, and JavaScript, along with an experience in full-stack development with React/Next.js and Flask, to contribute to the creation of innovative and impactful software solutions.

Education

Rochester Institute of Technology, Master of Science, Computer Science

Expected Dec. 2025

- GPA: 3.8/4.0

- Coursework: AI, ML, Algorithms, Advanced Databases: SQL/NoSQL, Computer Vision, Neural Networks, Information Retrieval

- **Research Assistant**

May 2025 - Present

- Engineered a Flask and JavaScript-based transcript analysis tool to help researchers analyze qualitative data, with features like audio processing, transcript management, demographic handling, and LLM-powered theme extraction.

- **Teaching Assistant**

Jan. 2025 - Present

- Supported graduate and undergraduate students in **C Programming** and both **introductory and advanced Big Data** courses by holding office hours, addressing Python and C-related queries, reviewing assignments and paper summaries, grading coursework, and providing personalized feedback on programming, data processing, and NoSQL concepts.

Dr. A.P.J. Abdul Kalam Technical University, Bachelor of Technology(Honors) Computer Science

July 2018 - July 2022

- GPA: 3.8/4.0

Skills

Languages: Python, Java, SQL, PostgreSQL, MongoDB, C/C++ , HTML/CSS, JavaScript, TypeScript, React

Libraries: Pandas, Numpy, Keras, Pytorch, Tensorflow, Scikit-learn, OpenCV

Technologies: Windows, Ubuntu Terminal, Linux, Spark, Hive, Hadoop, AWS Services – S3, Lambda, Glue

Developer Tools: Jupyter Notebook, Google Colab, PyCharm, IntelliJ, Git, Microsoft Office, PowerBI, Flask, Next.js

Experience

Sentari AI

Remote, USA

Intern

July 2025 - Sep 2025

- Collaborated in a fast-paced, agile team to design and build a full-stack AI application, using AI-assisted tools like Cursor to enhance productivity and contribute to team design reviews.
- Engineered a secure Supabase backend and a responsive UI with React/Next.js, implementing AI features such as Whisper transcription and GPT-powered insight generation.

Cognizant Technology Solutions Pvt Ltd (CTS)

Noida, India

Programmer Analyst

Feb. 2022 - June 2023

- Managed a cross-functional team of 15 peers across 4+ cities to design and implement a data warehouse using HDFS, Hadoop, and Hive, ensuring efficient distributed storage and enabling comprehensive data insights for business strategy.
- Advanced to Programmer Analyst within three months by optimizing ETL pipelines during Bayer's cloud transition project, effectively supporting data migration from Talend to Amazon Cloud to drive informed business decisions.

Antrix Academy (Froyo Technologies)

Noida, India

Intern

Sep. 2021 - Oct. 2021

- Applied agile methodologies to deliver end-to-end data science solutions—cleaned, mined, modeled, and visualized 10+ datasets using Python and Excel, implementing data strategy for optimized decision-making and client insights.

Projects

Real-Time ASL Word Recognition

Jan. 2025 - May 2025

- Developed a real-time deep learning model for American Sign Language (ASL) word recognition using MediaPipe Hands and GRU/LSTM networks for dynamic sequence modeling.
- Enhanced model robustness and integrated a novel live transcription pipeline using OpenCV, applying advanced regularization techniques to improve real-world performance.

Brain Tumor Detection

Aug 2024 - Dec 2024

- Developed a custom CNN that outperformed pre-trained models (DenseNet201, VGG16) in brain tumor classification, achieving 98.5% accuracy on 3,264 MRI images through extensive hyperparameter tuning and data augmentation.

Hybrid Approach for Automated Comment Updates in Code

Jan. 2024 - May 2024

- Devised a hybrid Seq2Seq and heuristic model for automated code comment updates, leveraging NLP techniques like tokenization and semantic similarity, achieving 30% higher accuracy significantly outperforming CUP and HebCUP.